Sample Solutions.

IT Maths Jan 2024

Question 1 (Compulsory): ALL TOPICS.
Attempt all parts.

a) Given the following matrices:

$$A = \begin{bmatrix} -2 \\ 6 \\ -1 \end{bmatrix} \quad B = \begin{bmatrix} 3 & 0 \\ 1 & -2 \\ 5 & 2 \end{bmatrix} \quad C = \begin{bmatrix} 5 & 1 & 3 \\ 3 & 2 & 7 \end{bmatrix}$$

i) size of each Matrix...

size of a Matrix = no. of rows(r) x no. of columns(c)

$$A = \begin{bmatrix} -2 \\ 6 \\ -1 \end{bmatrix} \begin{matrix} \text{row 1} \\ \text{row 2} \\ \text{row 3} \end{matrix}$$
col 1

size A = r x c
= 3 x 1

$$B = \begin{bmatrix} 3 & 0 \\ 1 & -2 \\ 5 & 2 \end{bmatrix} \begin{matrix} \text{row 1} \\ \text{row 2} \\ \text{row 3} \end{matrix}$$
col 1 col 2

size B = 3 x 2

$$C = \begin{bmatrix} 5 & 1 & 3 \\ 3 & 2 & 7 \end{bmatrix} \begin{matrix} \text{row 1} \\ \text{row 2} \end{matrix}$$
col 1 col 2 col 3

size C = 2 x 3.

ii) value of the elements ... .

element a_{ij} ← matrix name

row, column.

a_{21} → element in Matrix A, row 2, col 1

$$A = \begin{bmatrix} -2 \\ \textcircled{6} \\ -1 \end{bmatrix}$$

row 2

col 1

$a_{21} = 6$

b_{22} → element in Matrix B, row 2, col 2.

$$B = \begin{bmatrix} 3 & 0 \\ 1 & \textcircled{-2} \\ 5 & 2 \end{bmatrix}$$

row 2

col 2

$b_{22} = -2$

C_{23} → element in Matrix C, row 2, col 3.

$$C = \begin{bmatrix} 5 & 1 & 3 \\ 3 & 2 & \textcircled{7} \end{bmatrix}$$

row 2

col 3

$C_{23} = 7$

iii) Find the matrix given by $2B^T - C$

- Find B^T * transposed i.e. rows become columns, columns become rows.

$$B^T = \begin{bmatrix} 3 & 1 & 5 \\ 0 & -2 & 2 \end{bmatrix}$$

- Find $2B^T$ i.e. multiply the matrix B^T by the scalar (2).

$$2\begin{bmatrix} 3 & 1 & 5 \\ 0 & -2 & 2 \end{bmatrix} = \begin{bmatrix} 2(3) & 2(1) & 2(5) \\ 2(0) & 2(-2) & 2(2) \end{bmatrix}$$

$$2B^T = \begin{bmatrix} 6 & 2 & 10 \\ 0 & -4 & 4 \end{bmatrix}$$

- $2B^T - C$

$$\begin{bmatrix} 6 & 2 & 10 \\ 0 & -4 & 4 \end{bmatrix} - \begin{bmatrix} 5 & 1 & 3 \\ 3 & 2 & 7 \end{bmatrix} = \begin{bmatrix} 6-5 & 2-1 & 10-3 \\ 0-3 & -4-2 & 4-7 \end{bmatrix}$$

$$2B^T - C = \begin{bmatrix} 1 & 1 & 7 \\ -3 & -6 & -3 \end{bmatrix}$$

b) Find the determinant of

$$A = \begin{bmatrix} 1 & 7 & 5 \\ 3 & 0 & 1 \\ 6 & 3 & 2 \end{bmatrix} \quad (3\times3 \text{ Matrix})$$

Note: if only asked for the determinant of a 3x3 (and not the inverse of a 3x3), can use the shorter method (by diagonals) - simpler method.

see Brightspace page + Khan academy examples if using the cofactor method.

Repeat the first 2 cols

$$A = \begin{array}{ccc|cc} 1 & 7 & 5 & 1 & 7 \\ 3 & 0 & 1 & 3 & 0 \\ 6 & 3 & 2 & 6 & 3 \end{array}$$

Det A = Determinant

$$= (1\times0\times2) + (7\times1\times6) + (5\times3\times3) - (7\times3\times2) + (1\times1\times3) + (5\times0\times6)$$

start at top left corner and take each diagonal.

start bottom left corner & take each diagonal

$$= (0) + (42) + (45) - (42) + (3) + (0)$$

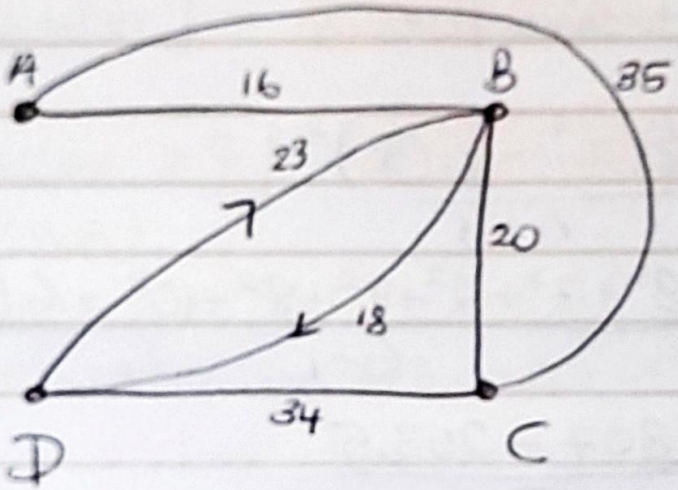
$$= 87 - 45$$

$$\det A = 42$$

c) i) Draw the graph of the given weight Matrix.

	A	B	C	D
A	0	16	35	∞
B	16	0	20	18
C	35	20	0	24
D	∞	23	34	0

18 direction!



ii) Directional edge = D to B = 23.

d) Recursion

$t(1) = 4$

$t(n) = t(n-1) + 2 \quad (n > 1)$

Replace n with 2 to get t(1) t(1) defined in definition

$t(2) = t(2-1) + 2$
$= t(1) + 2$
$= 4 + 2$
$t(2) = 6$

$t(3) = t(3-1) + 2$
$= t(2) + 2$ we know t(2) from previous step.
$= 6 + 2$
$t(3) = 8$

$t(4) = t(4-1) + 2$
$= t(3) + 2$
$= 8 + 2$ we know t(3) from previous step.
$t(3) = 10$

e) 3, 7, 2, 9, 8, 10

i) Mean $\bar{x} = \frac{\sum x_i}{n} = \frac{3+7+2+9+8+10}{6} = \frac{39}{6}$

$\bar{x} = 6.5$

ii) Standard deviation:
2 Formula to choose from.

$$S^2 = \frac{\sum x_i^2 - n(\bar{x})^2}{n-1}$$

$$= \frac{3^2+7^2+2^2+9^2+8^2+10^2 - 6(6.5)^2}{6-1}$$

$$= \frac{307 - 253.5}{5}$$

$S^2 = 10.7$

$S = \sqrt{10.7}$

$S = 3.27$ to 2 decimal places.

OR

$$S = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$$

$3 - 6.5 = (-3.5)^2 = 12.25$
$7 - 6.5 = (0.5)^2 = 0.25$
$2 - 6.5 = (-4.5)^2 = 20.25$
$9 - 6.5 = (2.5)^2 = 6.25$
$8 - 6.5 = (1.5)^2 = 2.25$
$10 - 6.5 = (3.5)^2 = 12.25$
53.5

$$= \sqrt{\frac{53.5}{5}}$$

$= \sqrt{10.7}$

$= 3.27$

$S = 3.27$

f) 12-sided dice...

i) Probability of an even number: P[even number]

even numbers: 2, 4, 6, 8, 10, 12 [ie 6 even numbers)
total number of possibilities = 12

$P[\text{even number}] = \frac{6}{12} = \frac{1}{2}$.

ii) $P[\text{number} > 9] = \frac{3}{12} = \frac{1}{4}$
(i.e. 10, 11, 12)

iii) P[even number or a number > 9]

let A = even number
B = a number > 9.

$P[A \text{ or } B] = P[A] + P[B] - P[A \text{ and } B]$.

Find $P[A \text{ and } B] = \frac{2}{12} = \frac{1}{6}$
i.e. (10, 12)

$\therefore P[A \text{ or } B] = \underbrace{\frac{6}{12}}_{\text{(part i)}} + \underbrace{\frac{3}{12}}_{\text{(part ii)}} - \frac{2}{12} = \frac{7}{12}$

$= \simeq 0.5833$

or ~58% chance.

g) calculate gradient (slope) of $y = 6x^2 - 3x + 1$ at the point (2,1)

- get dy/dx using power rule of differentiation

$$dy/dx = 12x - 3$$

$y = x^n$

$dy/dx = nx^{n-1}$

(multiply by the power)
(subtract 1 from the power)

slope at the point (2,1) — x, y

sub $x = 2$ into dy/dx.

$\therefore$ gradient (slope) $= 12(2) - 3$

$= 24 - 3$

$= 21$

h) Indefinite integral. use the power rule of integration.

$$\int (x^n).dx = \frac{x^{n+1}}{n+1} + C$$

(add 1 to the power, divide by the new power, add the constant of integration C)

i) $\int (8x^3 + 3x^2 - 6x + 7).dx$

$= \frac{8x^4}{4} + \frac{3x^3}{3} - \frac{6x^2}{2} + 7x + C$

$= 2x^4 + x^3 - 3x^2 + 7x + C$

ii) $\int (x^{-2}).dx = \frac{x^{-2+1}}{-2+1} + C$

$= \frac{x^{-1}}{-1} + C = -\frac{1}{x} + C$